

**Ex. No. : 07**

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## **Telephony Services**

### **Aim**

Develop an android application to display the information of the telephony services.

### ***Procedure:***

#### **Step 1 : File → New Project**

Provide the application name (e.g., "TelephonyServices") and click "Next".

#### **Step 2 : Select the target Android devices**

Select the minimum SDK to run the application. Click "Next".

#### **Step 3 : Choose the activity for the application**

By default, choose "Blank Activity". Click "Next".

#### **Step 4 : Enter activity name and click "Finish".**

#### **Step 5 : Edit the program**

Request necessary permissions Request in the manifest file.

Use Kotlin code in MainActivity.kt to implement telephony features ‘

such as dialing a number or retrieving SIM info

#### **Step 6: Run the application**

Two ways to run the application:

1. Running through emulator (with SD card configured)
2. Running through mobile device

## *AndroidManifest.xml*

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.telephony services">

    <!-- Telephony permissions -->
    <uses-permission android:name="android.permission.READ_PHONE_STATE" />
    <uses-permission android:name="android.permission.READ_PHONE_NUMBERS" />

    <!-- Feature requirements -->
    <uses-feature android:name="android.hardware.telephony"
        android:required="false" />

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.TelephonyServices">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

## *Activity\_main.xml*

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Telephony Services"
        android:textSize="22sp"
        android:textStyle="bold"
        android:textColor="@android:color/white"
        android:padding="16dp"
        android:background="#6200EE" />

    <!-- Network Operator Name -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Network Operator Name"
        android:textSize="14sp"
        android:paddingStart="16dp"
        android:paddingTop="8dp"
        android:textColor="#757575"/>

    <TextView
        android:id="@+id/networkOperatorNameTextView"
        android:layout_width="match_parent"
```

```

        android:layout_height="wrap_content"
        android:background="@drawable/text_view_border"
        android:padding="16dp"
        android:textSize="16sp"
        android:textColor="#212121"
        tools:text="Android" />

<!-- Phone Type -->
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Phone Type"
    android:textSize="14sp"
    android:paddingStart="16dp"
    android:paddingTop="8dp"
    android:textColor="#757575"/>

<TextView
    android:id="@+id/phoneTypeTextView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="@drawable/text_view_border"
    android:padding="16dp"
    android:textSize="16sp"
    android:textColor="#212121"
    tools:text="GSM" />

<!-- Network Country ISO -->
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Network Country ISO"
    android:textSize="14sp"
    android:paddingStart="16dp"
    android:paddingTop="8dp"
    android:textColor="#757575"/>

```

```
<TextView
    android:id="@+id/networkCountryIsoTextView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="@drawable/text_view_border"
    android:padding="16dp"
    android:textSize="16sp"
    android:textColor="#212121"
    tools:text="US" />
```

```
<!-- SIM Country ISO -->
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="SIM Country ISO"
    android:textSize="14sp"
    android:paddingStart="16dp"
    android:paddingTop="8dp"
    android:textColor="#757575"/>
```

```
<TextView
    android:id="@+id/simCountryIsoTextView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="@drawable/text_view_border"
    android:padding="16dp"
    android:textSize="16sp"
    android:textColor="#212121"
    tools:text="US" />
```

```
<!-- Device Software Version -->
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Device Software Version"
    android:textSize="14sp"
    android:paddingStart="16dp"
```

```
        android:paddingTop="8dp"
        android:textColor="#757575"/>
```

```
<TextView
```

```
    android:id="@+id/deviceSoftwareVersionTextView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="@drawable/text_view_border"
    android:padding="16dp"
    android:textSize="16sp"
    android:textColor="#212121"
    tools:text="---" />
```

```
<View
```

```
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1" />
```

```
<Button
```

```
    android:id="@+id/getServicesButton"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Get Telephony Services"
    android:textColor="@android:color/white"
    android:backgroundTint="#6200EE"
    android:padding="16dp"
    android:layout_margin="16dp"/>
```

```
</LinearLayout>
```

## *MainActivity.kt*

```
package com.example.telephony services

import android.Manifest
import android.content.Context
import android.content.pm.PackageManager
import android.os.Bundle
import android.telephony.TelephonyManager
import android.widget.Button
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AlertDialog
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
import androidx.core.content.ContextCompat

class MainActivity : AppCompatActivity() {

    private lateinit var networkOperatorNameTextView: TextView
    private lateinit var phoneTypeTextView: TextView
    private lateinit var networkCountryIsoTextView: TextView
    private lateinit var simCountryIsoTextView: TextView
    private lateinit var deviceSoftwareVersionTextView: TextView
    private lateinit var getServicesButton: Button

    private val TELEPHONY_PERMISSION_CODE = 101

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        // Initialize UI components
```

```

        networkOperatorNameTextView =
findViewById(R.id.networkOperatorNameTextView)

        phoneTypeTextView = findViewById(R.id.phoneTypeTextView)

        networkCountryIsoTextView =
findViewById(R.id.networkCountryIsoTextView)

        simCountryIsoTextView = findViewById(R.id.simCountryIsoTextView)

        deviceSoftwareVersionTextView =
findViewById(R.id.deviceSoftwareVersionTextView)

        getServicesButton = findViewById(R.id.getServicesButton)

// Set click listener for button
getServicesButton.setOnClickListener {
    checkTelephonyPermission()
}
}

private fun checkTelephonyPermission() {
    if (ContextCompat.checkSelfPermission(
        this,
        Manifest.permission.READ_PHONE_STATE
    ) != PackageManager.PERMISSION_GRANTED
    ) {
        // Show permission explanation dialog
        showPermissionDialog()
    } else {
        // Permission already granted, get telephony info
        getTelephonyInfo()
    }
}

private fun showPermissionDialog() {
    AlertDialog.Builder(this)
        .setTitle("Allow Telephony Services")

```



```

        .setMessage("to make and manage phone calls?")
        .setIcon(R.drawable.ic_phone)
        .setPositiveButton("ALLOW") { _, _ ->
            // Request permission
            ActivityCompat.requestPermissions(
                this,
                arrayOf(Manifest.permission.READ_PHONE_STATE),
                TELEPHONY_PERMISSION_CODE
            )
        }
        .setNegativeButton("DENY") { dialog, _ ->
            dialog.dismiss()
            Toast.makeText(
                this,
                "Permission denied. Cannot access telephony services.",
                Toast.LENGTH_SHORT
            ).show()
        }
        .create()
        .show()
    }

    private fun getTelephonyInfo() {
        val telephonyManager = getSystemService(Context.TELEPHONY_SERVICE) as
        TelephonyManager

        // Get network operator name
        val networkOperatorName = if (ActivityCompat.checkSelfPermission(
            this,
            Manifest.permission.READ_PHONE_STATE
        ) == PackageManager.PERMISSION_GRANTED
        ) {

```

```

        telephonyManager.networkOperatorName.isEmpty { "Android" }
    } else {
        "Permission required"
    }
    networkOperatorNameTextView.text = networkOperatorName

    // Get phone type
    val phoneType = when (telephonyManager.phoneType) {
        TelephonyManager.PHONE_TYPE_GSM -> "GSM"
        TelephonyManager.PHONE_TYPE_CDMA -> "CDMA"
        TelephonyManager.PHONE_TYPE_SIP -> "SIP"
        TelephonyManager.PHONE_TYPE_NONE -> "NONE"
        else -> "UNKNOWN"
    }
    phoneTypeTextView.text = phoneType

    // Get network country ISO
    val networkCountryIso =
        telephonyManager.networkCountryIso.uppercase().isEmpty { "US" }
    networkCountryIsoTextView.text = networkCountryIso

    // Get SIM country ISO
    val simCountryIso = telephonyManager.simCountryIso.uppercase().isEmpty
    { "US" }
    simCountryIsoTextView.text = simCountryIso

    // Get device software version
    val deviceSoftwareVersion = if (ActivityCompat.checkSelfPermission(
        this,
        Manifest.permission.READ_PHONE_STATE
    ) == PackageManager.PERMISSION_GRANTED
    ) {

```

```

        telephonyManager.deviceSoftwareVersion ?: "----"
    } else {
        "----"
    }

    deviceSoftwareVersionTextView.text = deviceSoftwareVersion

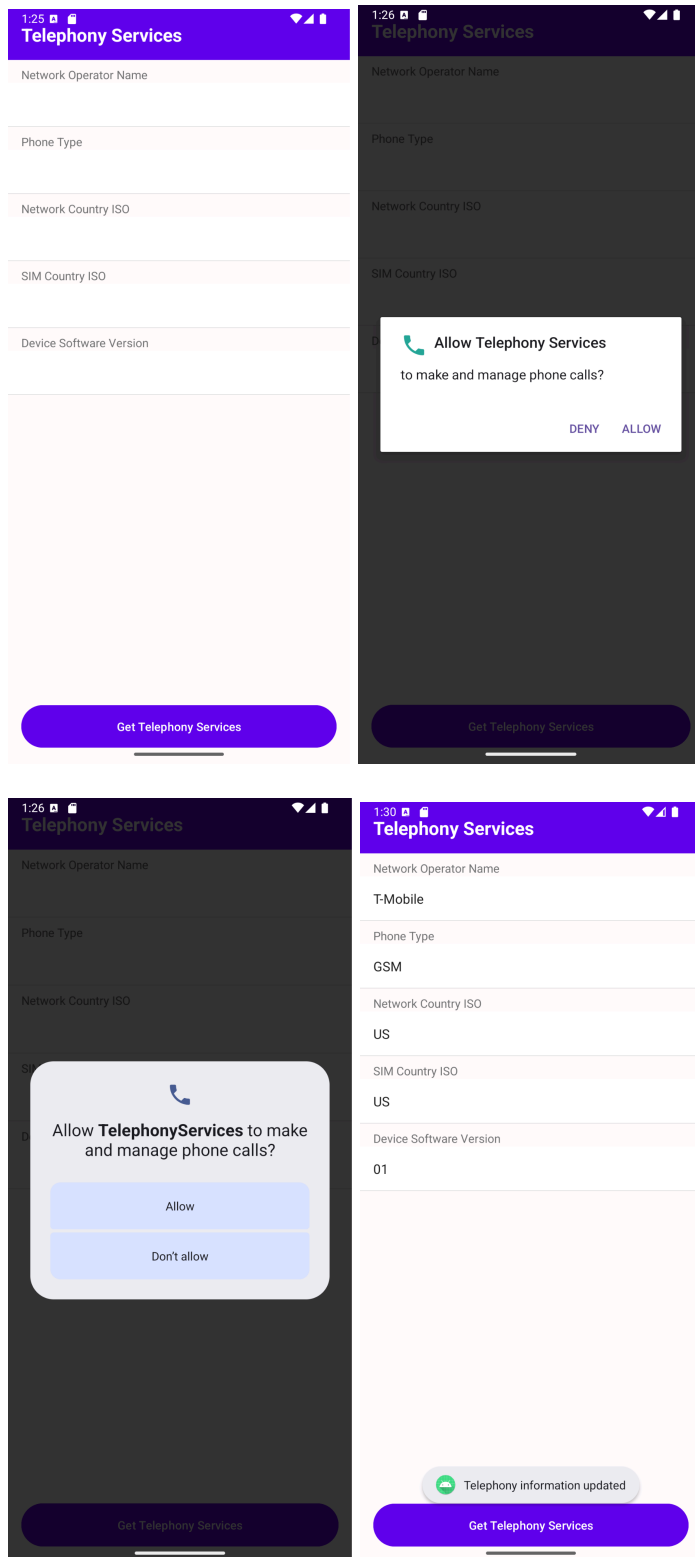
    Toast.makeText(this, "Telephony information updated",
        Toast.LENGTH_SHORT).show()
}

override fun onRequestPermissionsResult(
    requestCode: Int,
    permissions: Array<String>,
    grantResults: IntArray
) {
    super.onRequestPermissionsResult(requestCode, permissions,
        grantResults)

    if (requestCode == TELEPHONY_PERMISSION_CODE) {
        if (grantResults.isNotEmpty() && grantResults[0] ==
            PackageManager.PERMISSION_GRANTED) {
            // Permission granted, get telephony info
            getTelephonyInfo()
        } else {
            Toast.makeText(
                this,
                "Permission denied. Cannot access telephony services.",
                Toast.LENGTH_SHORT
            ).show()
        }
    }
}
}

```

## Output



**Result:**

The Telephony Services application successfully accesses telephony features like initiating phone calls or displaying SIM and network information on a mobile device with granted permissions.